CANDIDATE – PLEASE NOTE!

PRINT your name below and return this booklet with the answer sheet. Failure to do so may result in disqualification.

FORM TP 2014054

TEST CODE **01212010**

MAY/JUNE 2014

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE® EXAMINATION

CHEMISTRY

Paper 01 – General Proficiency

1 hour 15 minutes

06 JUNE 2014 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 15 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

   Sample Item

   The SI unit of length is the

   (A) metre
   (B) newton
   (C) second
   (D) kilogram

   Sample Answer

   ● (B) (C) (D)

   The best answer to this item is "metre", so answer space (A) has been shaded.
5. If you want to change your answer, erase it completely before you fill in your new choice.
6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to this item later. Your score will be the total number of correct answers.
7. You may do any rough work in this booklet.
8. Figures are not necessarily drawn to scale.
9. You may use a silent, non-programmable calculator to answer items.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

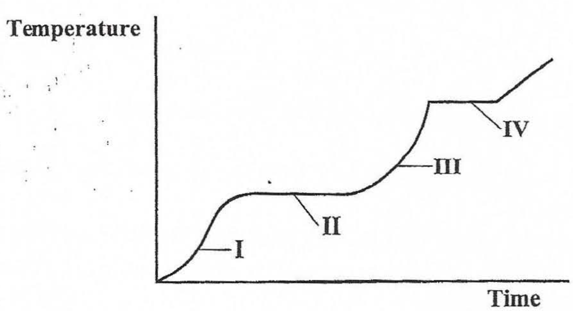
1. Which of the following movements depicts Brownian motion?

(A) Pollen grains moving at random in water
(B) Water moving from sea water into brine through cellophane
(C) The scent of perfume moving through the air in a room
(D) The movement of sugar particles into the blood stream through the intestine wall

2. The arrangements of electrons in atoms of X and Y are 2, 8, 5 and 2, 8, 6 respectively. Which of the following options represents X and Y?

	X	Y
(A)	Metal	Nonmetal
(B)	Nonmetal	Nonmetal
(C)	Nonmetal	Metal
(D)	Metal	Metal

Item 3 refers to the following graph which shows the changes in temperature with time, for a substance that is heated until no further physical change takes place.



3. During which portion of the curve is the substance a liquid only?

(A) I
(B) II
(C) III
(D) IV

4. The mass number of an element is the number of particles in the nucleus of an atom of that element. 'Particles' refer to

(A) neutrons
(B) protons and electrons
(C) neutrons and protons
(D) protons, neutrons and electrons

5. Which of the following elements has 7 electrons in its outer shell?

(A) Hydrogen
(B) Oxygen
(C) Nitrogen
(D) Chlorine

6. In the equation

$$x\,KOH + 3Br_2 \rightarrow y\,KBr + KBrO_3 + 3H_2O,$$

the values of x and y respectively are

(A) x = 5; y = 6
(B) x = 6; y = 5
(C) x = 5; y = 5
(D) x = 6; y = 6

7. Of the following ions, the one which requires the LARGEST number of moles of electrons to liberate one mole of it is

(A) Al^{3+}
(B) Cl^{-}
(C) Cu^{2+}
(D) O^{2-}

8. The quantity '1 mole of atoms of an element', refers to the mass of

(A) 1 atom of the element
(B) the element which contains 6.0×10^{23} atoms
(C) the element which occupies $24.0\ dm^3$ at STP
(D) the element which combines completely with 12 g of carbon –12

9. Which of the following substances does NOT have a giant structure?

(A) Copper
(B) Graphite
(C) Oxygen
(D) Sodium chloride

10. Which of the following statements BEST describes the formation of a metallic bond? A metallic bond is formed when

(A) anions are held together by negative electrons
(B) metal atoms are held together by molecular forces
(C) cations are held together by a sea of mobile electrons
(D) positive metal ions are held together by a sea of anions

11. From which of the following substances can a solid be obtained by the process of sedimentation?

(A) Gels
(B) Foams
(C) Emulsions
(D) Suspensions

12. The mass concentration of a potassium chloride solution is 60 g dm^{-3}. What is the mass of potassium chloride in 25 cm^3 of this solution?

(A) 0.0015 g
(B) 0.15 g
(C) 1.5 g
(D) 15.0 g

13. Separation of mixtures is based on specific properties of the components in the mixture. Which of the following properties is used to separate a mixture of oil and water?

(A) Solubility
(B) Viscosity
(C) Particle size
(D) Boiling point

14. Which of the following separation techniques is NOT used during the extraction of sucrose from sugar cane?

(A) Filtration
(B) Precipitation
(C) Centrifugation
(D) Chromatography

15. Barium is below calcium in Group II of the Periodic Table. When these metals react with water, the main differences in observation is the

(A) gas evolved
(B) rate of reaction
(C) colour of the product
(D) colour change with litmus

16. Graphite can be used as a lubricant because of the

(A) loose electrons which can move throughout the lattice
(B) weak attraction among the hexagonal layers of carbon atoms
(C) strong attraction among the hexagonal layers of carbon atoms
(D) strong attraction within the hexagonal layers of carbon atoms

17. Which of the following halogens is a liquid at room temperature?

(A) Bromine
(B) Fluorine
(C) Chlorine
(D) Iodine

18. Which of the following terms BEST describes strong acids?

(A) Concentrated
(B) Partially ionized
(C) Ionized to a large extent
(D) Concentrated and partially ionized

19. A solution has a pH of 1. This solution would be expected to react with

(A) zinc metal to produce hydrogen
(B) zinc metal to produce a solution of pH 10
(C) hydrochloric acid to produce chlorine
(D) hydrochloric acid to produce a salt and water

Item 20 refers to one mole of EACH of the following acids.

I. H_2SO_4
II. CH_3COOH
III. $(COOH)_2$

20. Which of the above acids would require MORE than one mole of NaOH(aq) for complete neutralization?

(A) II only
(B) I and III only
(C) II and III only
(D) I, II and III

21. Which of the following terms BEST describes the oxide of a metal?

(A) Salt
(B) Base
(C) Alkali
(D) Acid

22. A solution, when reacted with the gas sulfur dioxide, becomes green. The solution contains potassium

(A) nitrate
(B) sulfate
(C) dichromate(VI)
(D) manganate(VII)

23. In the electrolysis of aqueous copper(II) sulfate solution using copper electrodes, the anode

(A) decreases in mass
(B) increases in mass
(C) undergoes no change in mass
(D) is the site where reduction occurs

24. The equation for the reaction of silicon with chlorine is

$Si(s) + 2Cl_2(g) \rightarrow SiCl_4(l)$.

In this reaction, silicon is the reducing agent and is

(A) oxidized with an increase in oxidation state
(B) oxidized with a decrease in oxidation state
(C) reduced with an increase in oxidation state
(D) reduced with a decrease in oxidation state

25. Which of the statements below is true for the following equation?

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$$

(A) Mg is reduced from oxidation state +2 to 0.
(B) Mg is oxidized from oxidation state 0 to +2.
(C) Hydrogen is reduced from oxidation state +2 to 0.
(D) Hydrogen is oxidized from oxidation state 0 to +1.

26. 2.32 g of an oxide of iron were heated in dry hydrogen. When the reaction was completed, 1.68 g of iron were formed.

[RAM: Fe = 56; O = 16]

The oxide of iron was

(A) reduced
(B) oxidized
(C) neutralized
(D) decomposed

27. Acidified potassium manganate(VII) is

(A) a reducing agent
(B) an oxidizing agent
(C) a dehydrating agent
(D) both an oxidizing agent and reducing agent

28. During the electrolysis of aqueous copper(II) sulfate solution using inert electrodes, the ions migrating to the cathode are

(A) Cu^{2+} (aq) and SO_4^{2-}(aq)

(B) H^+(aq) and OH^-(aq)

(C) SO_4^{2-}(aq) and OH^-(aq)

(D) Cu^{2+} (aq) and H^+ (aq)

Items 29–30 refer to the following options. Each option may be used once, more than once or not at all to answer the items below.

(A) Iron
(B) Lead
(C) Copper
(D) Magnesium

Which element

29. forms an oxide which cannot be reduced by hydrogen?

30. would be liberated from an aqueous solution of its salts during electrolysis?

31. Which of the following statements about electrolysis is NOT correct?

(A) The electrons leave the solution via the positive electrode.
(B) The electrons enter the solution via the negative electrode.
(C) Decomposition of the electrolyte at the electrode produces an electric current.
(D) Decomposition of the electrolyte at the electrode is due to an electric current.

32. How many coulombs (C) of electricity pass through an electrolyte when a current of 100 A flows for 10 minutes?

(A) 100
(B) 600
(C) 1 000
(D) 60 000

33. The rate of a chemical reaction does NOT depend on the

(A) presence of a catalyst
(B) concentration of the reactants
(C) energy change of the overall reaction
(D) temperature of a reacting system

34. At a flour mill, there is the risk of an explosive reaction occurring when a match is struck. This is due MAINLY to the

(A) increased pressure in the area
(B) increased concentration of flour
(C) high temperature caused by the lighted match
(D) large surface area created by the large number of small particles

35. Which of the following metals can be extracted by electrolysis?

(A) Zn
(B) Ca
(C) Al
(D) Fe

36. When crystals of potassium nitrate are dissolved in water, the temperature of the solution decreases because

(A) energy is required to break down the crystal structure of the potassium nitrate
(B) heat is always absorbed when a substance dissolves
(C) the water absorbs the energy from the potassium nitrate
(D) the energy content of dissolved potassium nitrate is lower than that of solid potassium nitrate

37. Which of the following pairs of substances is MOST suitable for producing carbon dioxide?

(A) Dilute hydrochloric acid and calcium carbonate
(B) Dilute hydrochloric acid and calcium hydroxide
(C) Sodium hydroxide and calcium carbonate
(D) Dilute sulfuric acid and calcium hydroxide

Items 38–39 refer to the following sequence of reactions involving iron compounds, where I, II, III and IV represent progressive stages in the sequence.

$$FeCO_3 \xrightarrow{\text{I}} FeO \xrightarrow{\text{II}} Fe_2O_3 \xrightarrow{\text{III}} Fe_2(SO_4)_3 \xrightarrow{\text{IV}} Fe(OH)_3$$

38. In which stage is the oxidation state of iron increased?

(A) I
(B) II
(C) III
(D) IV

39. A suitable reagent that could be used at IV is

(A) steam
(B) hydrogen
(C) solid copper oxide
(D) aqueous sodium hydroxide

---

40. Which of the following metals can displace zinc from a solution of zinc sulfate?

(A) Iron
(B) Lead
(C) Copper
(D) Aluminium

Items 41–42 refer to the properties of four metals, W, X, Y and Z, which are summarised in the following table.

Metal	Reaction With Water/Steam	Products Formed When Nitrate is Heated
W	There is no reaction.	Metal, NO_2 and O_2
X	Reacts with steam but not with cold water.	Oxide, NO_2 and O_2
Y	Reacts rapidly with cold water.	Nitrite and oxygen
Z	Reacts slowly with cold water.	Oxide, NO_2 and O_2

41. The order of reactivity of these metals, with the MOST reactive FIRST, is

(A) W, X, Y, Z
(B) X, Y, Z, W
(C) Y, Z, W, X
(D) Y, Z, X, W

42. Which of these metals forms a carbonate which is stable to heat?

(A) W
(B) X
(C) Y
(D) Z

43. Which of the following substances forms dense white fumes with ammonia gas?

(A) Oxygen
(B) Hydrogen
(C) Nitrogen dioxide
(D) Hydrogen chloride

44. Which of the following observations is likely when EXCESS aqueous ammonia is added to a solution of copper(II) sulfate and the mixture is shaken?

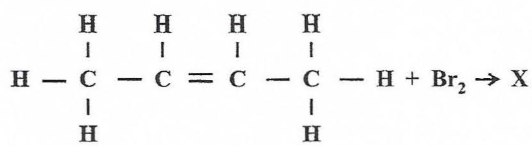
(A) Green solution
(B) Green precipitate
(C) Deep blue solution
(D) Light blue precipitate

45. Which of the following molecules could be obtained by cracking the alkane, C_8H_{18}?

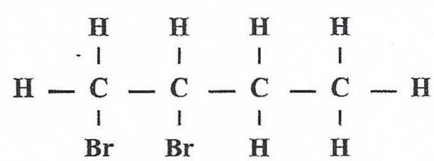
I. C_2H_4
II. C_6H_{14}
III. $C_{16}H_{34}$

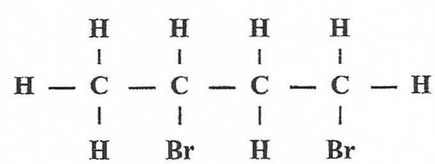
(A) I only
(B) I and II only
(C) II and III only
(D) I, II and III

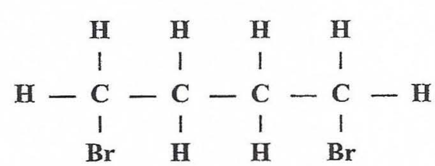
Item 46 refers to the following equation

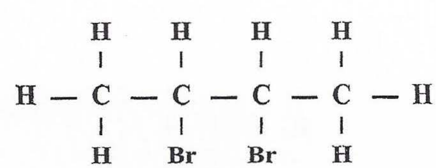


46. What is the correct structural formula of X?

(A) 

(B) 

(C) 

(D) 

47. Which of the following substances is a product of a halogenation reaction with methane?

(A) Ammonia
(B) Water
(C) Hydrogen chloride
(D) Nitrogen(IV) oxide

Items **48–49** refer to the following options.

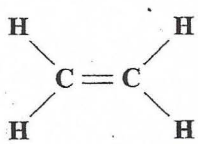
(A) C_2H_4
(B) C_2H_6
(C) $CH_3(CH_2)_2COOH$
(D) $CH_3(CH_2)_3CH_2OH$

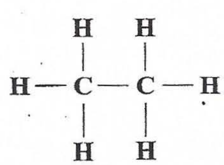
Match EACH item below with ONE of the options above. Each option may be used once, more than once, or not at all.

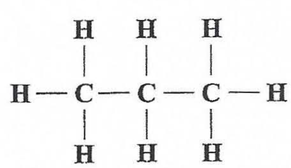
48. An alcohol

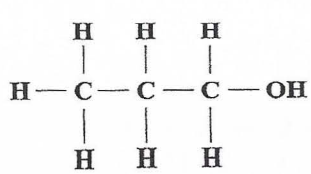
49. An alkane

50. Which of the following compounds can form a polymer?

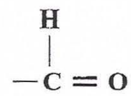
(A)


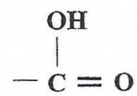
(B)


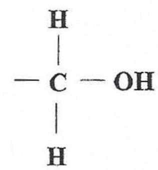
(C)


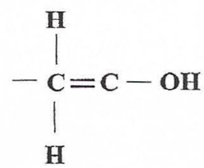
(D)


51. The functional group present in carboxylic acids is

(A)


(B)


(C)


(D)


Items **52–55** refer to the following options.

(A) Bitumen
(B) Paraffin oil
(C) Methane
(D) Propene

Match EACH item below with ONE of the above options. Each option may be used once, more than once or not at all.

52. The residue in the fractional distillation of petroleum

53. Is a MAJOR constituent of natural gas

54. Is used in making plastics

55. Requires the LEAST amount of oxygen per mole for complete combustion

56. A polyamide is formed by a condensation reaction between a molecule containing

(A) at least one -COOH group with another containing at least one -NH_2 group
(B) at least one -OH group with another containing at least one -COOH group
(C) at least one -OH group with another containing at least one -NH_2 group
(D) two -OH groups with another containing two -COOH groups

57. The compound, ethene, is described as being unsaturated. This means that the

(A) carbon atoms in ethene are linked by single bonds
(B) carbon atoms in the molecule are very reactive
(C) molecule has insufficient hydrogen atoms
(D) molecule contains at least one double bond

58. Nylon is a macromolecule which is held together by the same linkage as a protein molecule. What is the name given to this type of linkage?

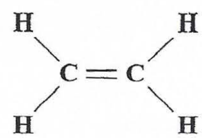
(A) Amide bond
(B) Double bond
(C) Ester linkage
(D) Ether linkage

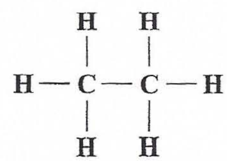
59. Which of the following statements is TRUE?

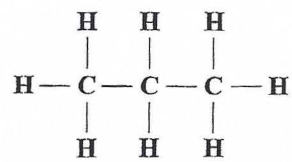
The particles in a gas

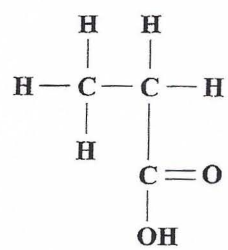
(A) are larger when heated
(B) are further apart when heated
(C) are very closely packed together
(D) have strong forces between them

60. Which of the following compounds can decolourize potassium manganate(VII)?

(A)


(B)


(C)


(D)


END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.